Robotic Technology & Application

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Catalog

Part 1. Underwater Robot

Part 2. Review of USV

Part 3. A novel method of USV autonomous cruise

Part 1

Underwater Robot

Chapter **25** of "*Springer Handbook of Robotics*"

1. Background

Underwater robots or **unmanned underwater vehicles** (UUVs) can help us better understand marine and other environmental issues, protect the ocean resources of the earth from pollution, and efficiently utilize them for human welfare.

Remotely operated vehicles (ROVs): tethered and remotely operated.

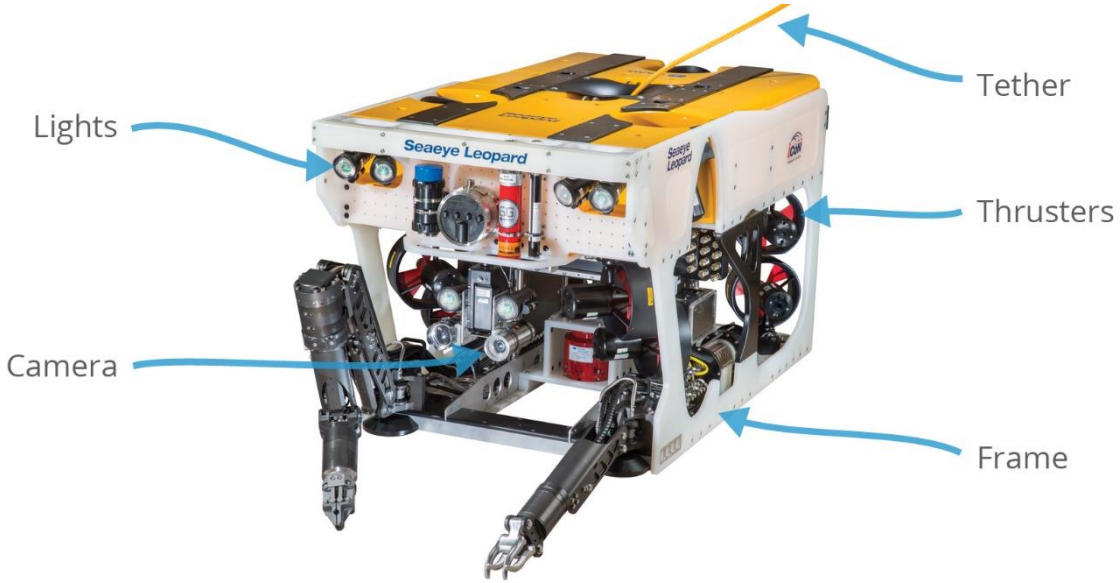


Fig.1 ROV

- Video cameras and lights
- Sonars
- Still camera
- Manipulator
- Grabber
- ……

1. Background

- 1960s: The US Navy funded the early ROV technology development, which were used to recover underwater ordnance.

- **Various applications**:
  - Military operations
  - Environmental systems
  - Scientific missions
  - Ocean mining
  - Gas and oil industry
  - ……



Fig.2 ROV recover underwater ordnance

1. Background

- **ROV**: very high operational costs with mother vessels and operator fatigue…
- **AUV (autonomous underwater vehicles)**

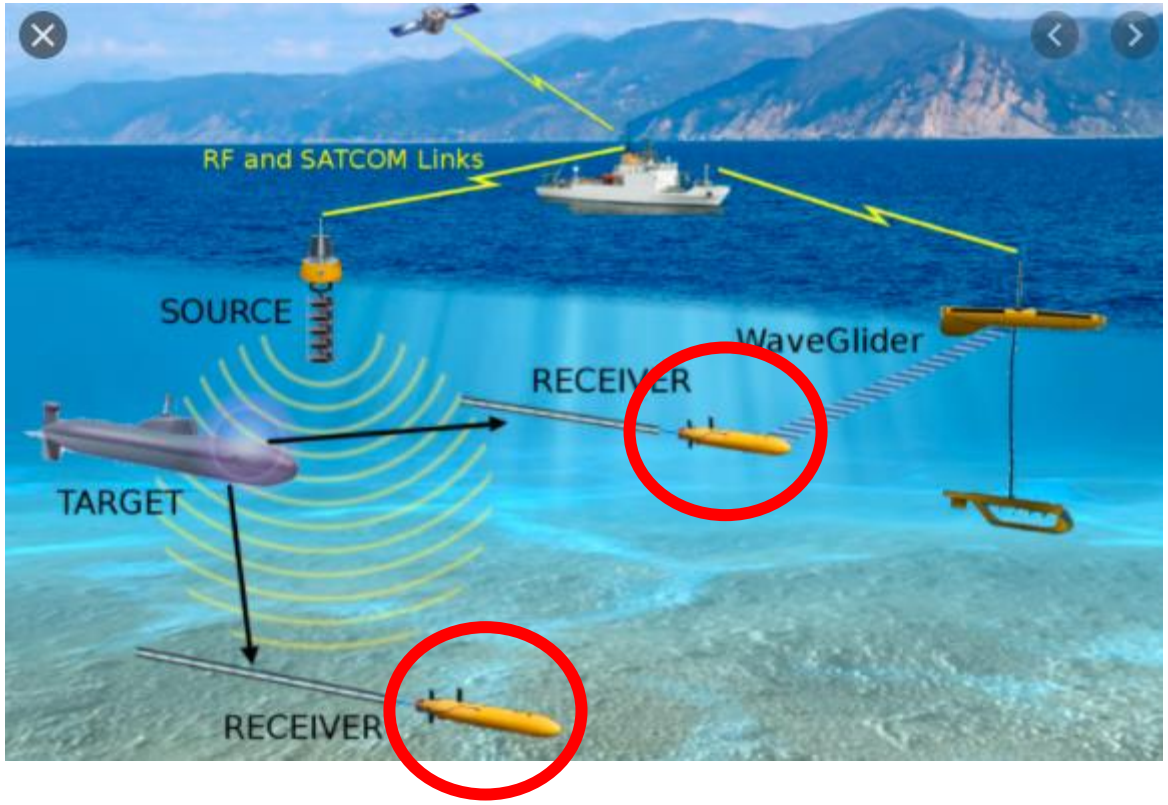


Fig.3 AUV

2. Mechanical Systems

- Every 10m ~ 1 atm of water pressure
- Chemical environment: highly corrosive

→

- Excellent structural strength
- Excellent chemical and environmental resistance

Mechanical Structure:

1. **Frame**:

   primary skeleton of the robot

2. **Pressure vessel**:

   a cylindrical or spherical shape is the most effective way to maximize strength at the expense of low space efficiency.

3. **Fairing**:

   skin of the robot

2. Mechanical Systems

The frame must be designed considering the following **operational conditions**:

1. **On the ground without motion**: static gravitational force
2. **Underwater in motion**:

   buoyancy force, gravitational force, all thruster forces in a dynamic state
3. **Hanging from a crane for launch and recovery**: enormous snap force

Requirement of frame and pressure vessel:

- High **strength** and high **stiffness**
- Good **corrosion resistance** and **machinability** in choosing **materials(Tab. 1)**

Requirement of buoyancy materials:

- Low **density**, high **strength**, **nontoxic** harmless, and **weather resistance**
- Low **water absorption** and low **compressibility**

3. Power Systems

Power:

- **On-board power**:
  - AIP system

Battery **installation**:
- Housed in a pressure vessel
- Using pressure-tolerant and waterproof batteries

- **Surface power by a tether**
  - A surface system: a shipboard power source, frequency converter, step-up transformers, and a ground fault monitoring device
  - A tether
  - A subsea system: step-down transformers, rectifiers, power busses, power switching devices for distribution, and a ground fault monitoring device.
- **The combination of the two**

3. Power Systems

- System configuration and setup reflects the **power requirement** of the devices installed on a ROV including thrusters, lights, sensors and any circuits. A detail configuration depends on the form of current and needed capacity.
- **Design objectives**:
  - Maximizing power **transfer efficiency** and **lifetime**
  - Minimizing the **overall system cost** using steady-state analysis and transient analysis
  - Minimize the diameter of a cable, system cost, support systems on deck, and the weight of the ROV without losing a high level of electrical power quality

Tab.2 Summary of pros and cons of DC and AC (alternating current)

Form	Pro	Con	Remarks
DC	Two conductors with less shielding in tether, low voltage drop	Current protection, difficulties in voltage step-up and down	For observation class ROV
AC	Longer transmission distances, flexibility in voltage up and down	Inductance noise, reactive losses, need subsystem for freq. converting	For deep-sea ROV

4. Underwater Actuators and Sensors

The overall performance depends on the quality of robot's actuators and sensors.

Actuator Classification:

- Thrusters → Mapping between thruster command and generated force is **not linear**.
- Water jets
- Biomimetic actuators

Sensor Classification:

- Navigation sensors
- System sensors
- Mission sensors

- ~~GPS, Electromagnetic wave propagation~~
- Acoustic-based sensing
- Nonacoustic-based sensing

4. Underwater Actuators and Sensors

Tab.3 Summary of different type of Actuators

Type	Efficiency	Application	Pros & Cons
Electrical Thrusters	43%	Small ROVs and AUVs	Compact size and high efficiency
Electro-hydraulic Thrusters	15%	Large size work-class ROVs	Extensive thrust and practicality in the rugged conditions
Bio-inspired smart actuators			
Jet propulsion			

- **Typical electrical thruster**:

  Electric motor, motor amplifier, controller, thruster housing, oil-filled, drive shafts.

4. Underwater Actuators and Sensors

Nonacoustic Sensors

- Magnetic sensors
- Inertial sensors
  - Accelermeters
  - Gyroscope(Gyro)
- Doppler velocity log
- **Pressure sensors**

Sensor Fusion

- **Attitude and heading reference system(AHRS)**
  - Gyros, Accelerometers, and Magnetometers on all three axes, and an onboard microprocessor running a nonlinear estimation algorithm
- **Inertia navigation system（INS）**
  - Accelerometers, Gyros, Navigation computer, and some electronics
- **Underwater computer vision system**
  - as part of navigation sensors for underwater robots in addition to simply recording images of the underwater environment
- **Advanced, high-quality camera and video management system**

Acoustic Sensors

- **Acoustic Doppler current profilers (ADCPs)**: water speed sensors
- **Sonar systems**
  - Single-beam directional sonar
  - Scanning sonar
  - Side scan sonar
  - Multibeam sonar
  - Interferometric sonar
  - Synthetic aperture sonar
- **Acoustic positioning systems** →
  - Long baseline (LBL)
  - Short baseline (SBL
  - Ultrashort baseline (USBL)
  - Interrogate
  - Responder
- **GPS intelligent buoys(GIB) system**

Other Sensor Issues

- **Multiple sonar sensors**
- **Health-monitoring sensors**

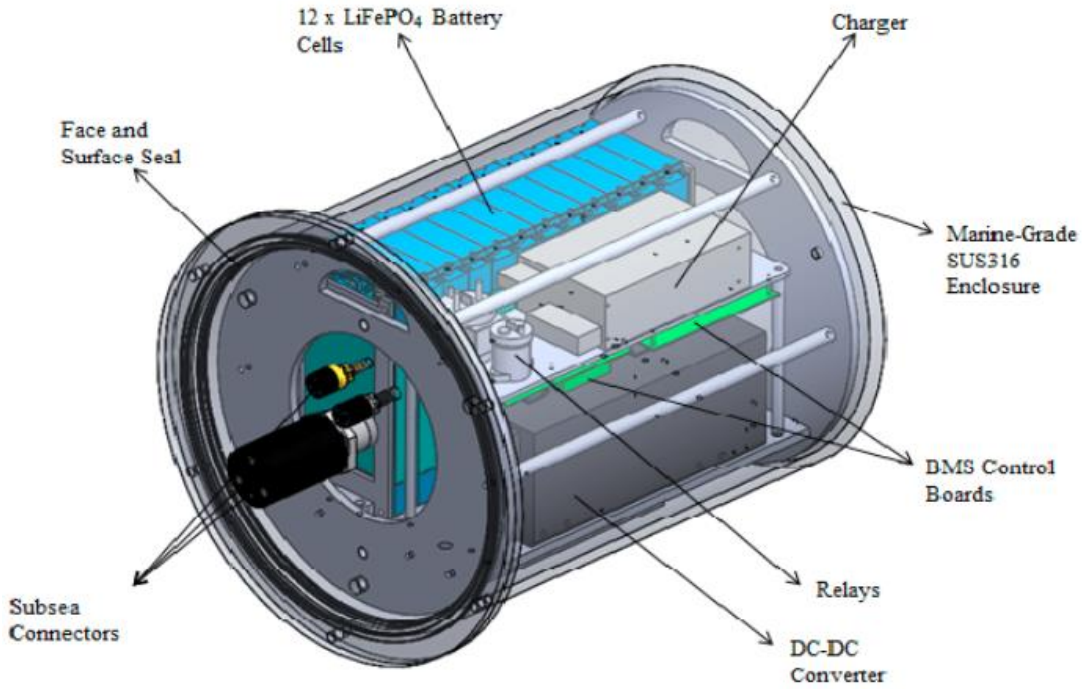
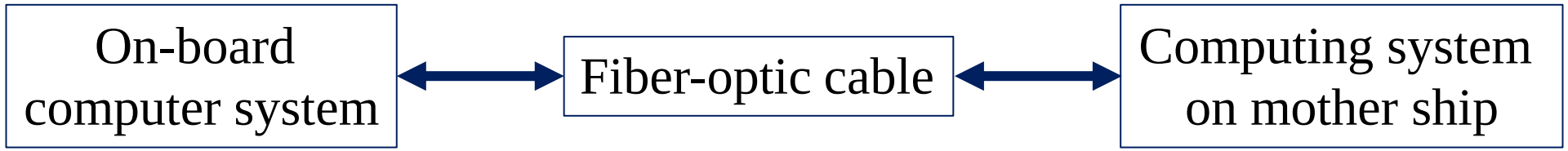


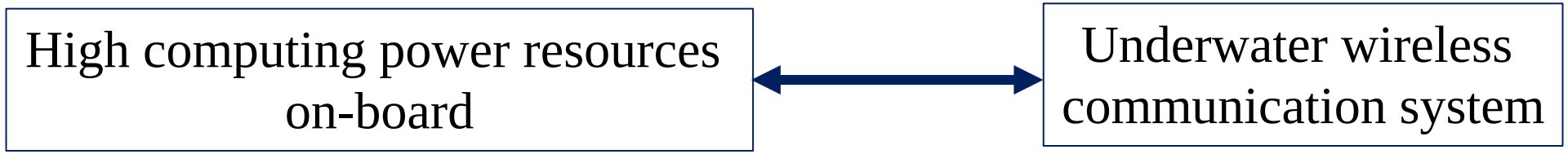
Fig.4 BMS

- **Emergency plan**

- **ROVs**


On-board computer system
Fiber-optic cable
Computing system on mother ship

- **AUVs**


High computing power resources on-board
Underwater wireless communication system

5. Computers, Communications, and Architecture

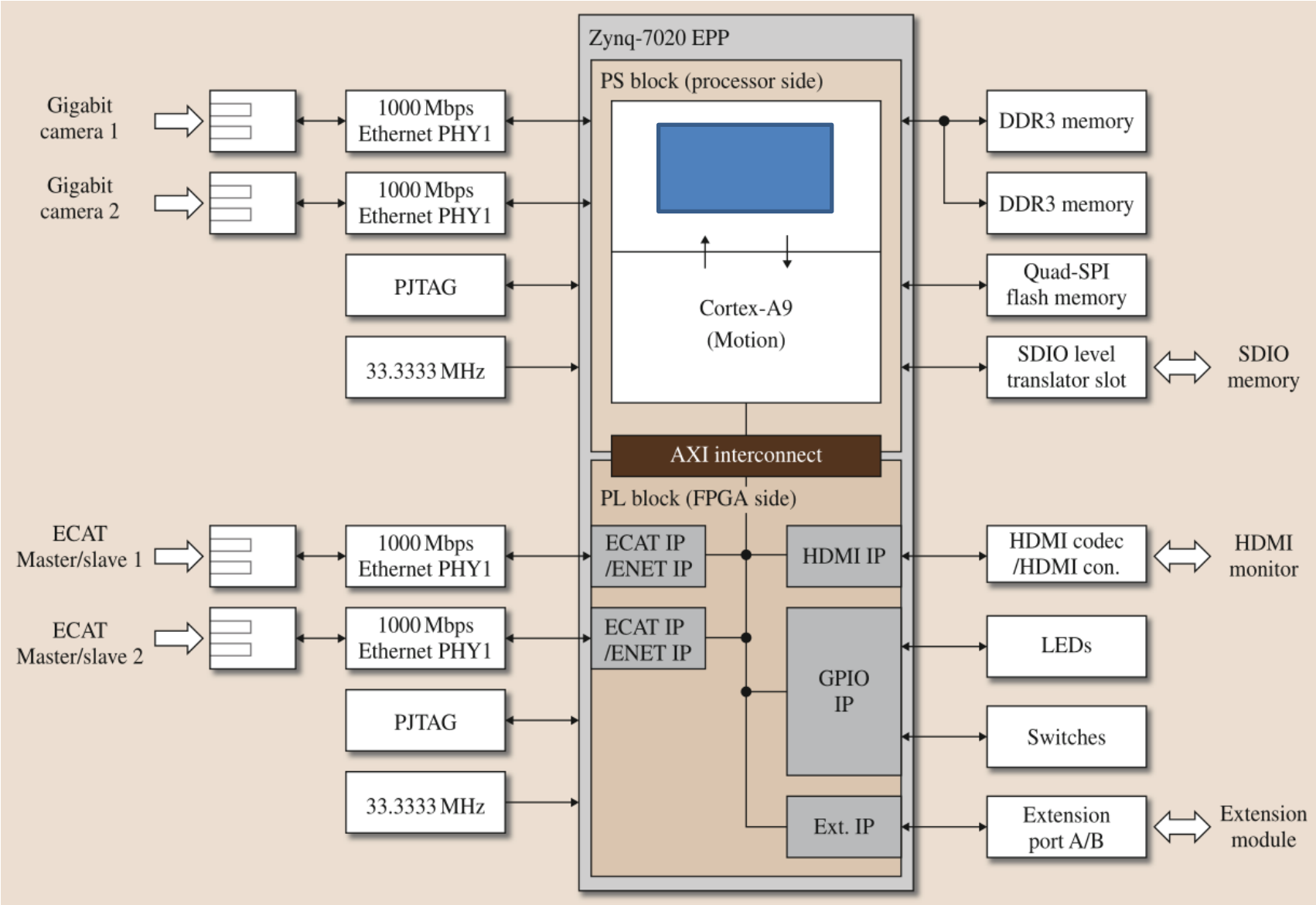


Fig.5 Hardware configuration of Zynq-7020-based system, mController

5. Computers, Communications, and Architecture

- **Radio communication**
  - on the surface
- **Acoustic communication**
  - underwater environment
- **Electromagnetic waves**
  - close-range communication
- **Optical underwater communication**
  - much higher data transfer rates than an acoustic communication system

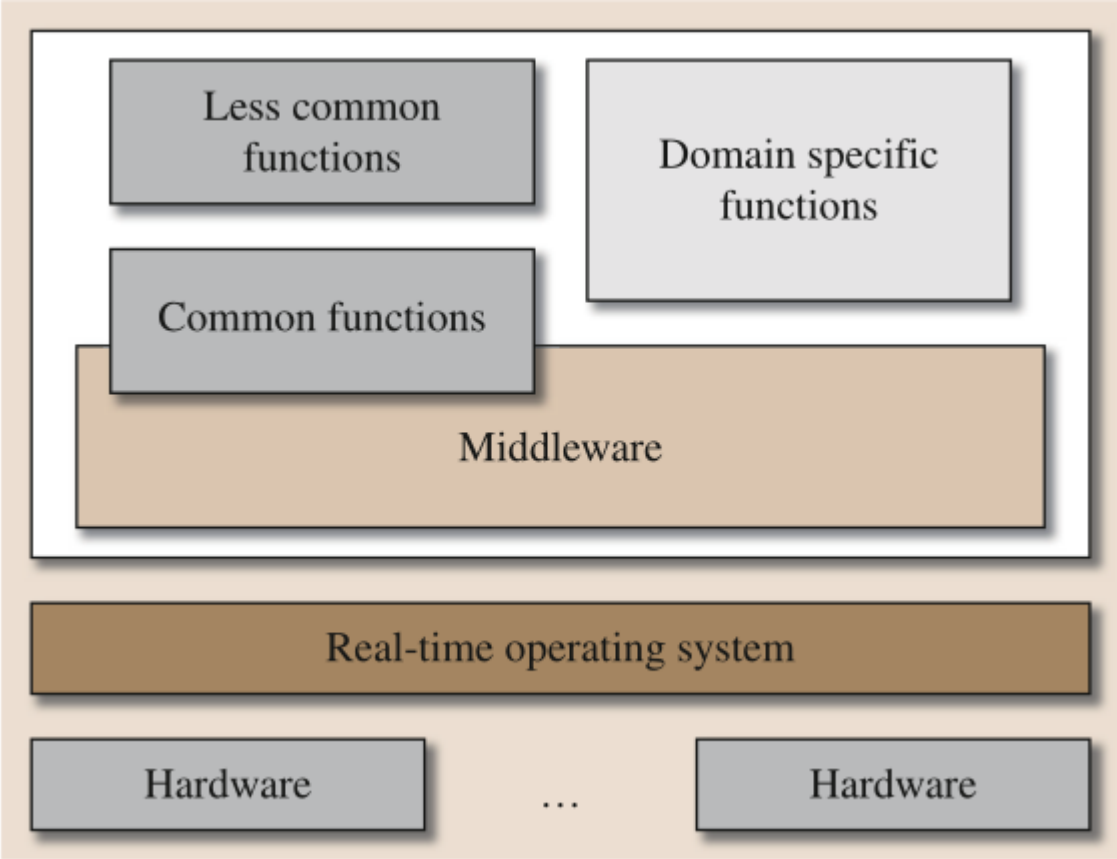


Fig.6 Typical configuration of software for robot

AUV architectures:

- Hierarchical architecture
- Hierarchical architecture
- Subsumption architecture
- Hybrid architecture

OR

- Deliberative architecture
- Reactive architecture
- Hybrid architecture

5. Computers, Communications, and Architecture

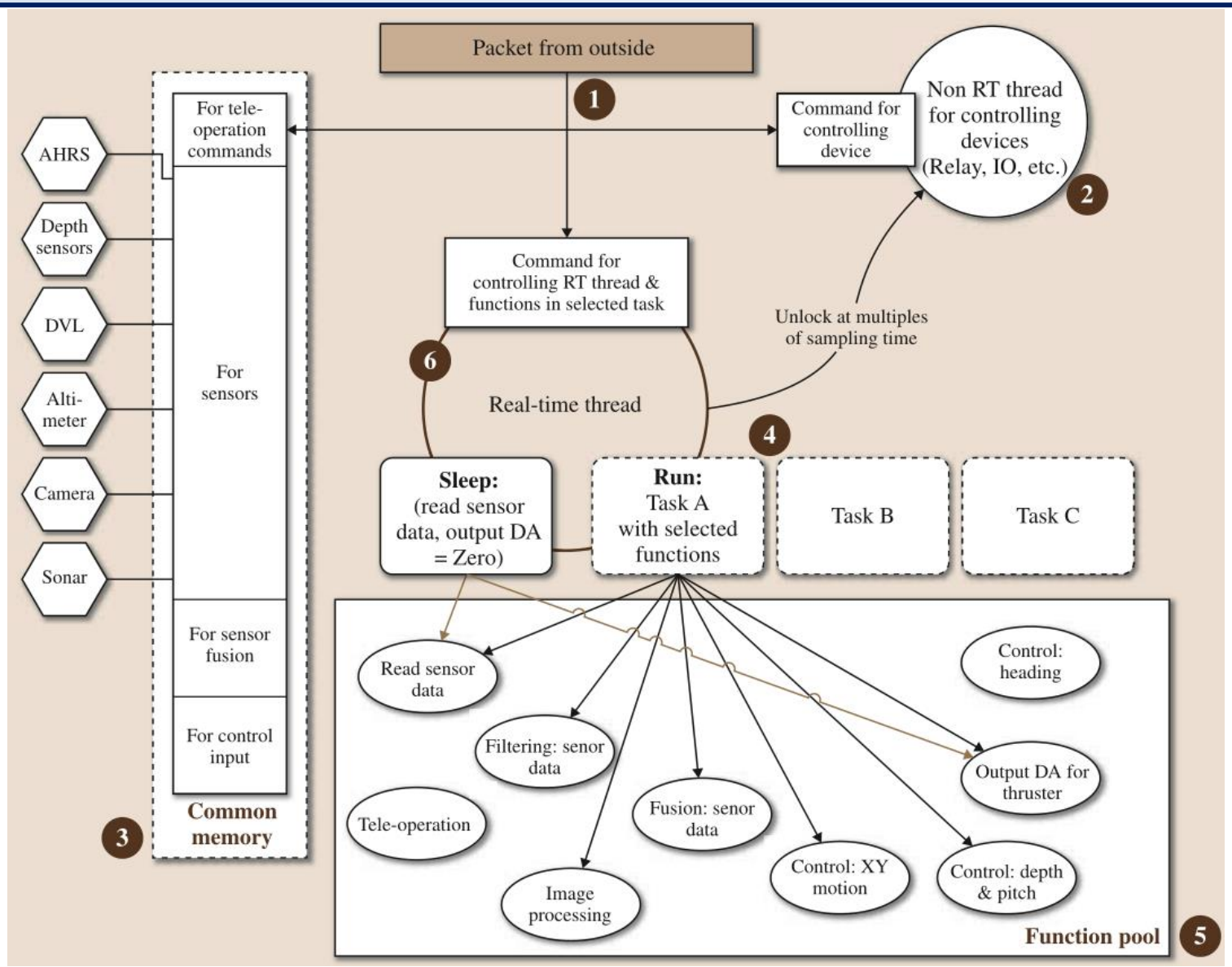


Fig.7 Example diagram of simple software architecture

6. Underwater Manipulators

- **Typical work-class ROV**:
  - A 7 DOF manipulator
  - A 5 DOF grabber
- **2 operator for intervention missions**:
  - Steering the vehicle
  - Operating the manipulator

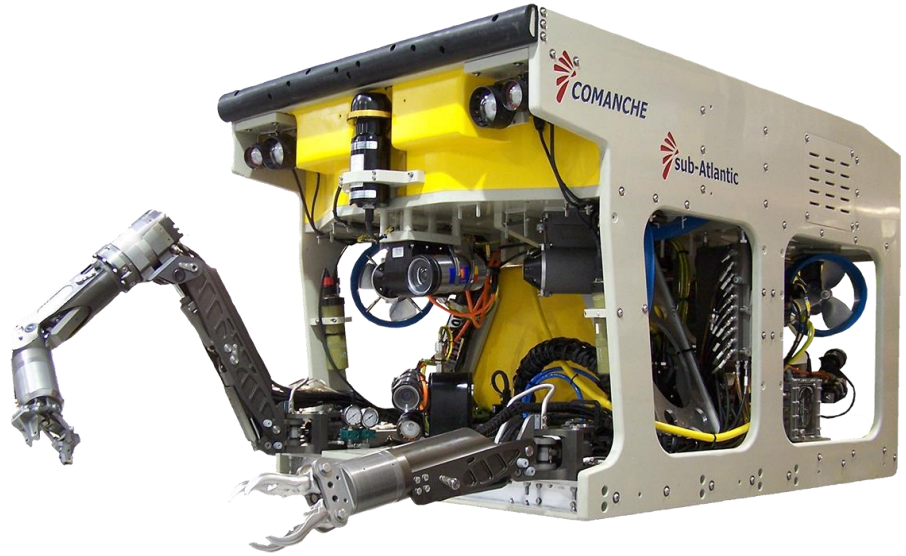


Fig.8 ROV with manipulator

- Very few **AUV** systems for intervention missions, instead of a test-bed for developing autonomous manipulation or as a working model for research.

7 Conclusions and Future

Major subsystems of underwater robots:

- Mechanical systems
- Power systems
- Actuators
- Sensors
- Computing systems
- Software architecture
- Communications
- Manipulator systems

Application:

- **ROV**
  - Scientific research
  - Military operations
  - Underwater constructions
- **AUV**
  - post-incident monitoring of BP Macondo oil leakage(2010)
  - recovery of black boxes of Air France flight 447(2009)

Part 2

Unmanned surface vehicles:
An overview of developments and challenges

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Review

Unmanned surface vehicles: An overview of developments and challenges

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Abstract

With growing worldwide interest in commercial, scientific, and military issues associated with both oceans and shallow waters, there has been a corresponding growth in demand for the development of **unmanned surface vehicles (USVs)** with advanced **guidance, navigation and control (GNC)** capabilities. This paper presents a comprehensive literature review of recent progress in USVs development. The paper first provides an overview of both historical and recent USVs development, along with some fundamental definitions. Next, existing USVs GNC approaches are outlined and classified according to various criteria, such as their applications, methodologies, and challenges. Finally, more general challenges and future directions of USVs towards more practical GNC capabilities are highlighted.

1. Introduction

Unmanned surface vehicles (USVs)

Autonomous surface vehicles (ASVs)

Autonomous surface crafts (ASCs)



Fig.2-1 Climate Change



Fig.2-3 USV

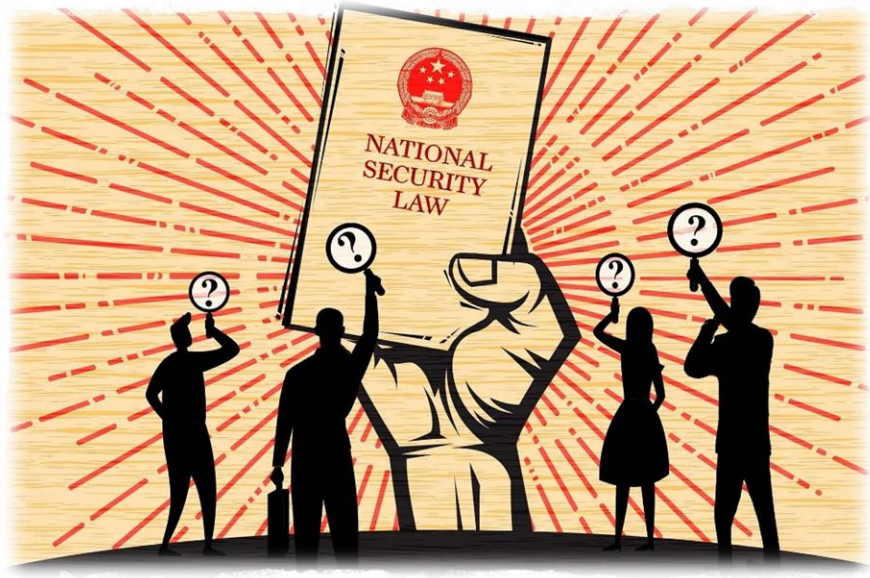


Fig.2-2 National Security Issue

1. Introduction

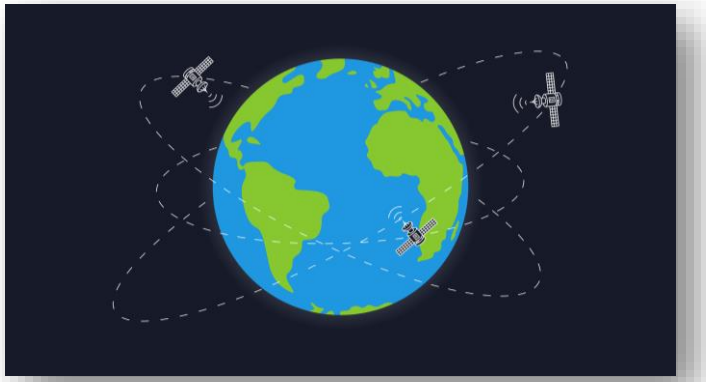


Fig.2-4 GPS

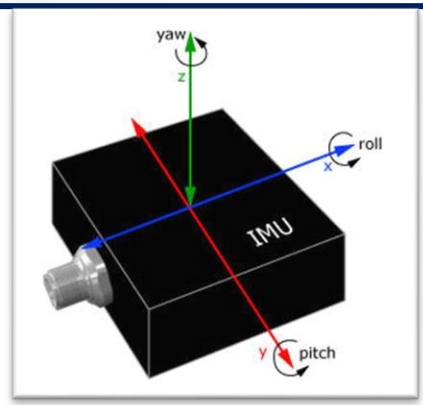


Fig.2-5 IMU

Tab.2-1 Potential applications of USVs

Types	Specific applications
Scientific research	Bathymetric survey (Roberts & Sutton, 2006); ocean biological phenomena, and migration and changes in major ecosystems (Goudey et al., 1998); ocean activities research; multi-vehicle cooperation (cooperative work among aerial, ground, water surface or underwater vehicles) (Majohr & Buch, 2006; Yan, Pang, Sun, & Pang, 2010); as experimental platforms for the purpose of testing hull designs, communication and sensor equipments, propulsion and operating systems, as well as control schemes (Breivik, 2010; Vaneck, Rodriguez-Ortiz, Schmidt, & Manley, 1996)
Environmental missions	Environmental monitoring, samplings, and assessment (Caccia et al., 2005; Naeem et al., 2008; Rasal, 2013; Svec et al., 2014b); disaster (like tsunami, hurricane, eruption of submarine volcano) aided prediction and management, and emergency response (Murphy et al., 2008); pollution measurements and clean-up
Ocean resource exploration	Oil, gas and mine explorations (Pastore & Djapic, 2010; Roberts & Sutton, 2006); offshore platform/pipeline construction and maintenance (Bertram, 2008; Breivik et al., 2008)
Military uses	Port, harbor, and coastal surveillance, reconnaissance and patrolling (Caccia et al., 2007; Kucik, 2004; Pastore & Djapic, 2010; Svec & Gupta, 2012); search and rescue (Murphy et al., 2008; Roberts & Sutton, 2006); anti-terrorism/force protection (Campbell et al., 2012); mine countermeasures (US Navy, 2007); remote weapons platform (Bertram, 2008); target drone boats (Roberts & Sutton, 2006)
Other applications	Transportation (Kiencke et al., 2006); mobile communication relays (Caccia et al., 2008a); refueling platform for USVs, unmanned aerial vehicles (UAVs), unmanned underwater vehicles (UUVs), and other manned vehicles

1. Introduction

Following advantages of USVs can be identified:

(1) USVs can perform **longer and more hazardous missions** than manned vehicles;

(2) maintenance costs are lower and **personnel safety** is far greater since no crew is onboard;

(3) the **low weight** and **compact dimensions** of USVs give them enhanced maneuverability and deployability in shallow waters (riverine and coastal areas) where larger craft cannot operate effectively;

(4) USVs also have greater potential payload capacity and are able to perform deeper water depth monitoring and sampling compared to other aircraft/UAVs and spacecraft.

2. State-of-the-art USV systems

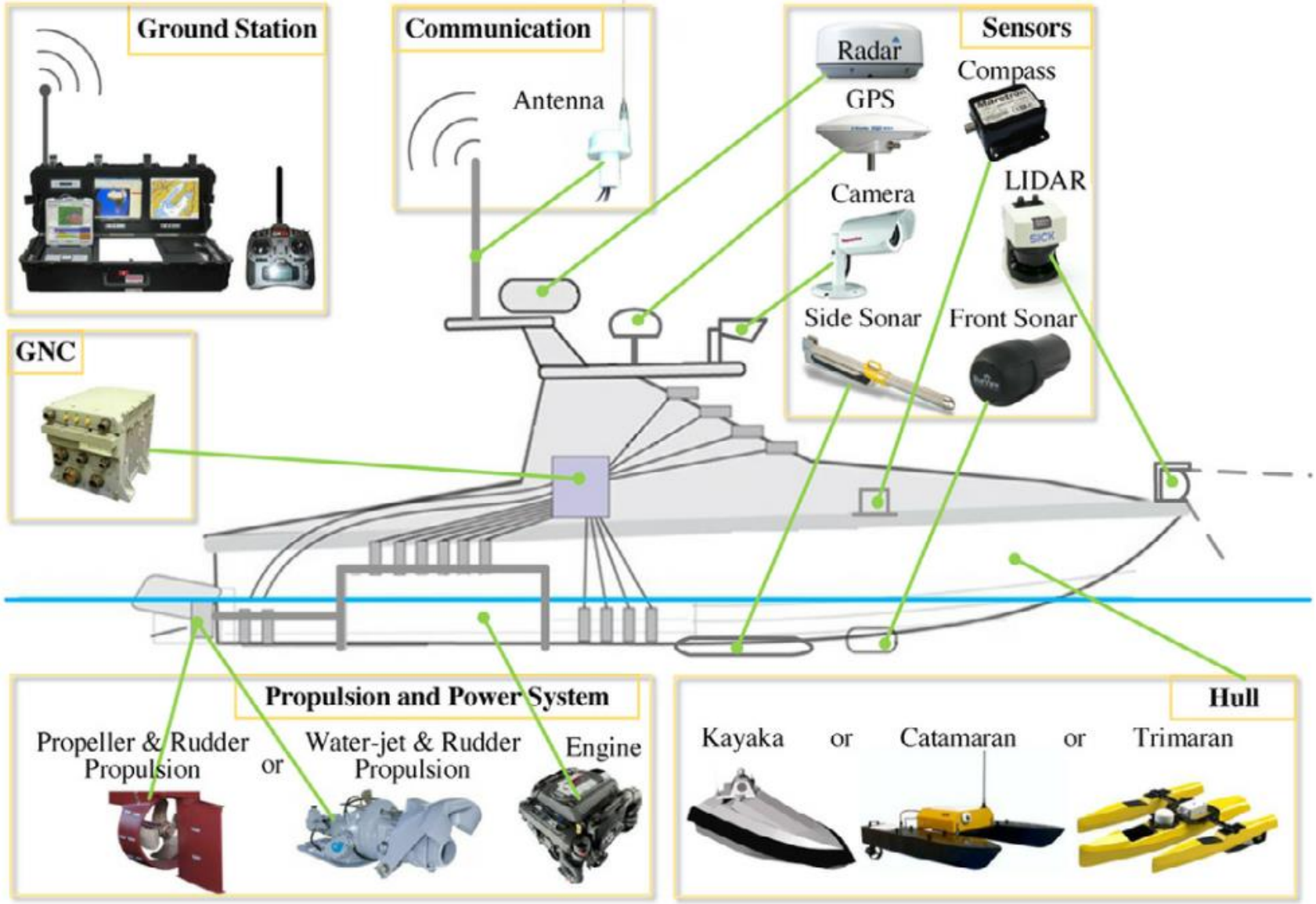


Fig.2-6 Fundamental architecture of a typical USV

2. State-of-the-art USV systems

USV subsystems

- Guidance system
- Navigation system
- Control system

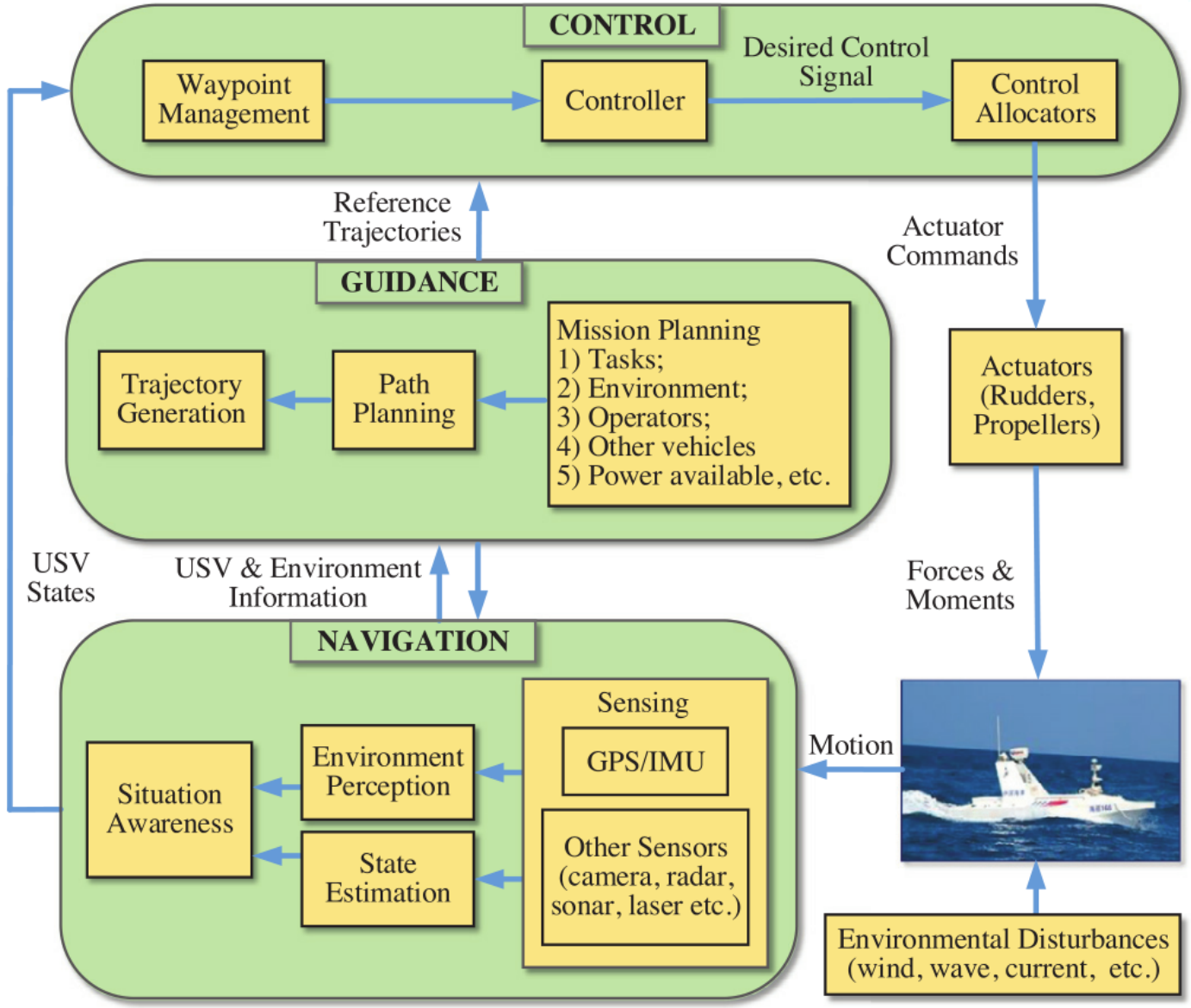
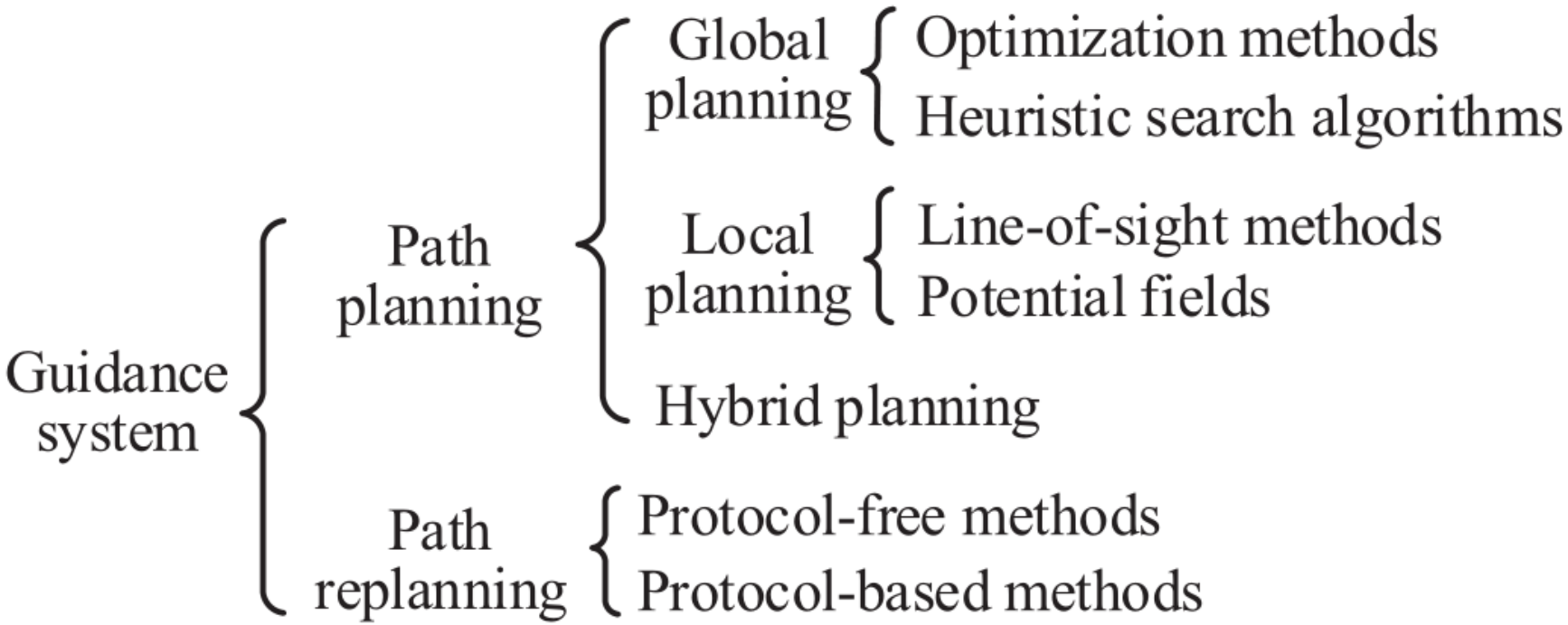
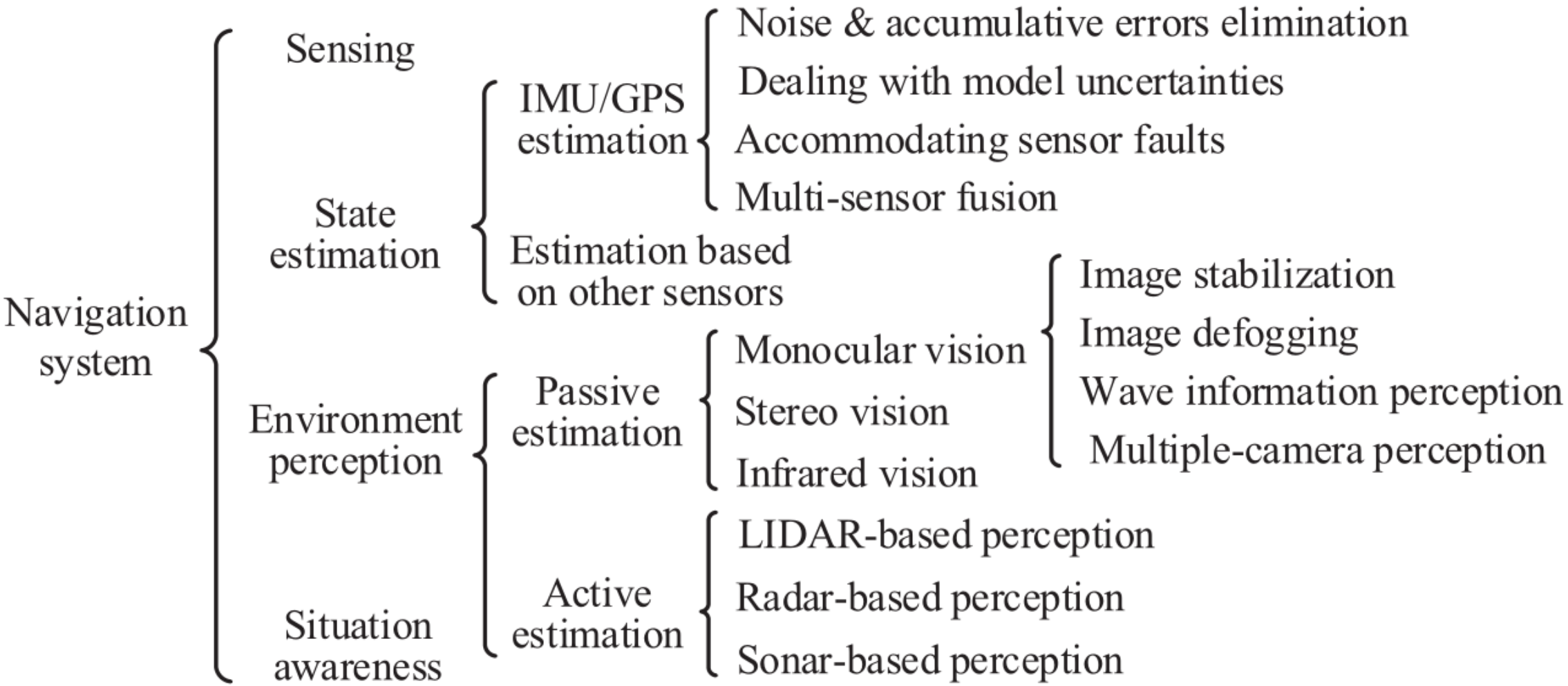


Fig.2-7 General structure of USV guidance, navigation, and control systems

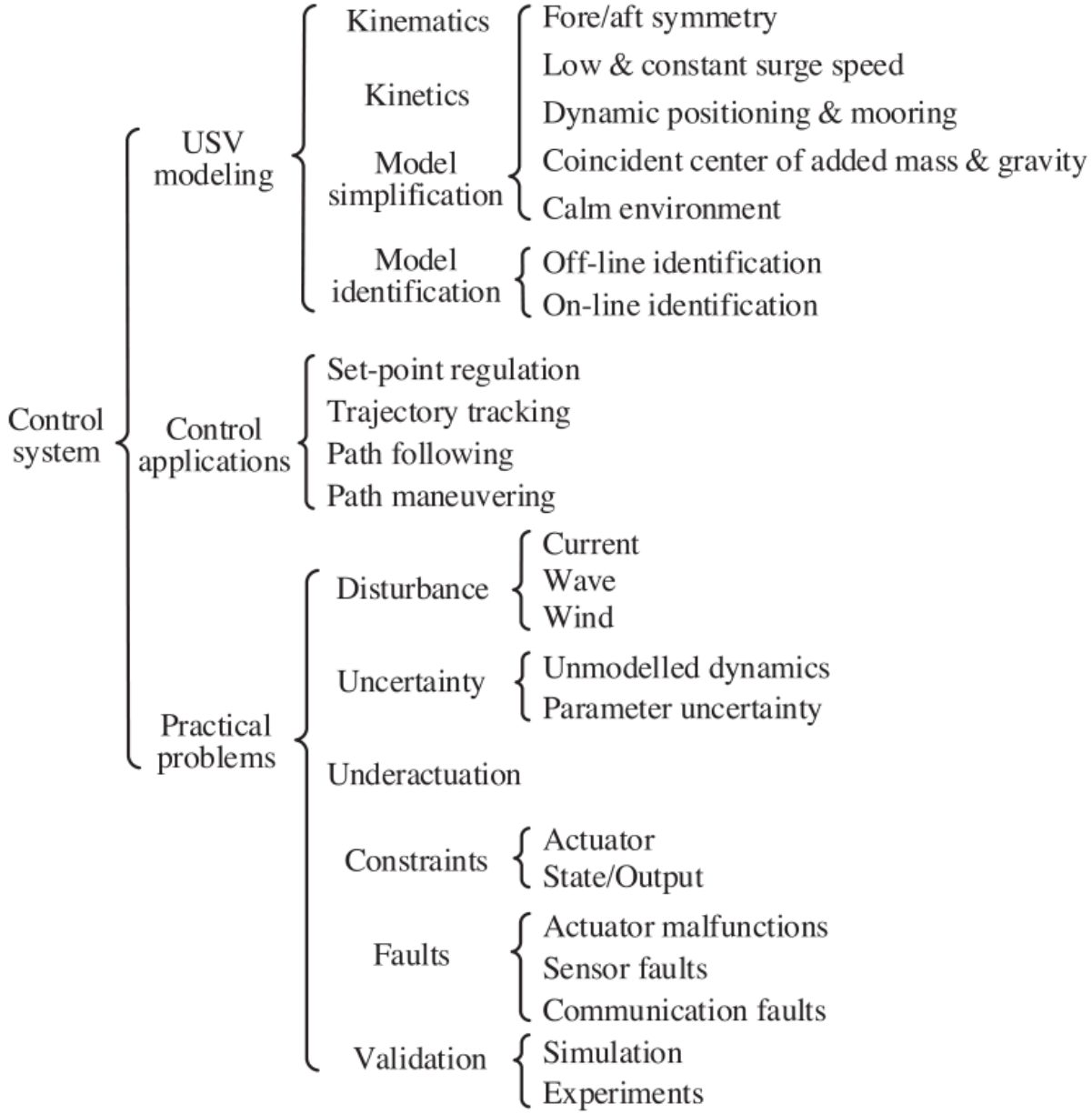
3. Classification of USV guidance techniques



4. Classification of USV navigation techniques



5. Classification of USV control techniques



6. Key GNC technologies for multiple USVs and other unmanned vehicles

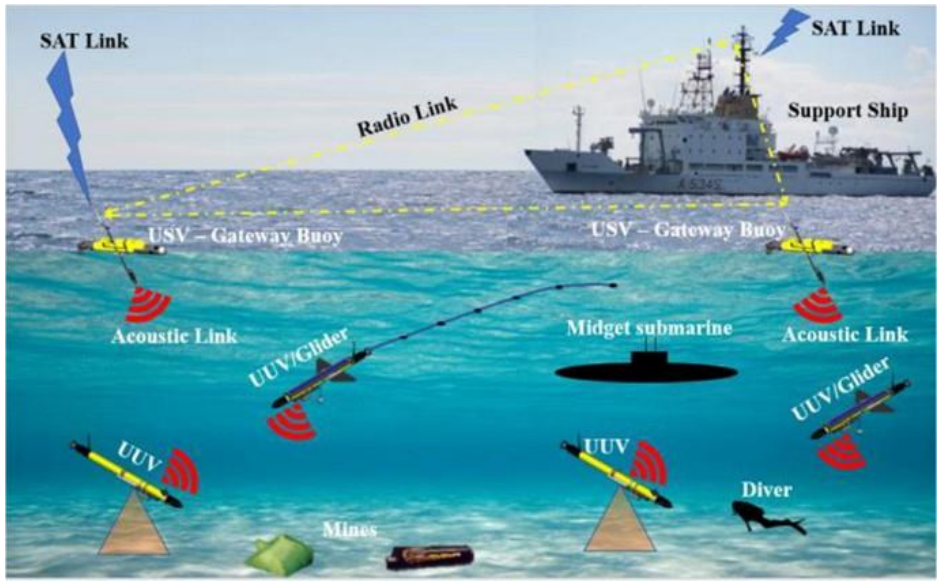


Fig.2-8 cooperation of USVs and other diverse vehicles

- Cooperation between USVs
- Cooperation with UAVs
- Cooperation with UUVs

Theoretical and practical challenges:

1. Disturbances and uncertainties
2. Communication limitations
3. Collision avoidance
4. Underway replenishment
5. Experimental validation

7. Challenges and future directions

- **Autonomous GNC of single USV**
  - Guidance
  - Navigation
  - Control
- **Autonomous GNC of USVs and other vehicles**
  - GNC in cooperative systems
  - Cooperation of USVs and other unmanned/manned Systems

8. Conclusions

In the near future, the development of **fully autonomous** USVs in highly dynamic maritime environments remains an open question, and there are numerous ongoing research works on this topic. This paper has presented a technical review and bibliographical list on historical and contemporary developments in USV GNC systems. The **basic definitions** of USVs system are given. The **adopted methodologies for USV GNC** are categorized and outlined. Some **challenges and future directions** have also been presented to facilitate the research progress of autonomous and practically applicable USVs.

Part 3

A novel method of unmanned surface vehicle autonomous cruise

Abstract

This paper aims to propose a new method for combining **global path planning** with **local path planning**, to provide an efficient solution for unmanned surface vehicle (USV) path planning despite the changeable environment.

A search of shortcut **Dijkstra** algorithm was used to control the USV in the global path planning. When the USV encounters unknown obstacles, it switches to our modified **artificial potential field (APF)** algorithm for local path planning.

The method in this paper offers a solution to the issue of path planning in changeable or unchangeable environment, and was confirmed by simulations and experiments. Both the simulation and experimental results demonstrate that the combinatorial path planning method is more efficient in the complex environment.

1. Introduction

To perform missions using unmanned surface vehicle (USV), **autonomous navigation** is a central capability.

However, most researches only focused on improving a single obstacle avoidance algorithm and neglected the combinatorial effectiveness of global obstacle avoidance and local obstacle avoidance.



Fig.3-1 The procedure of the combinatorial method

2. Search of shortcut Dijkstra algorithm

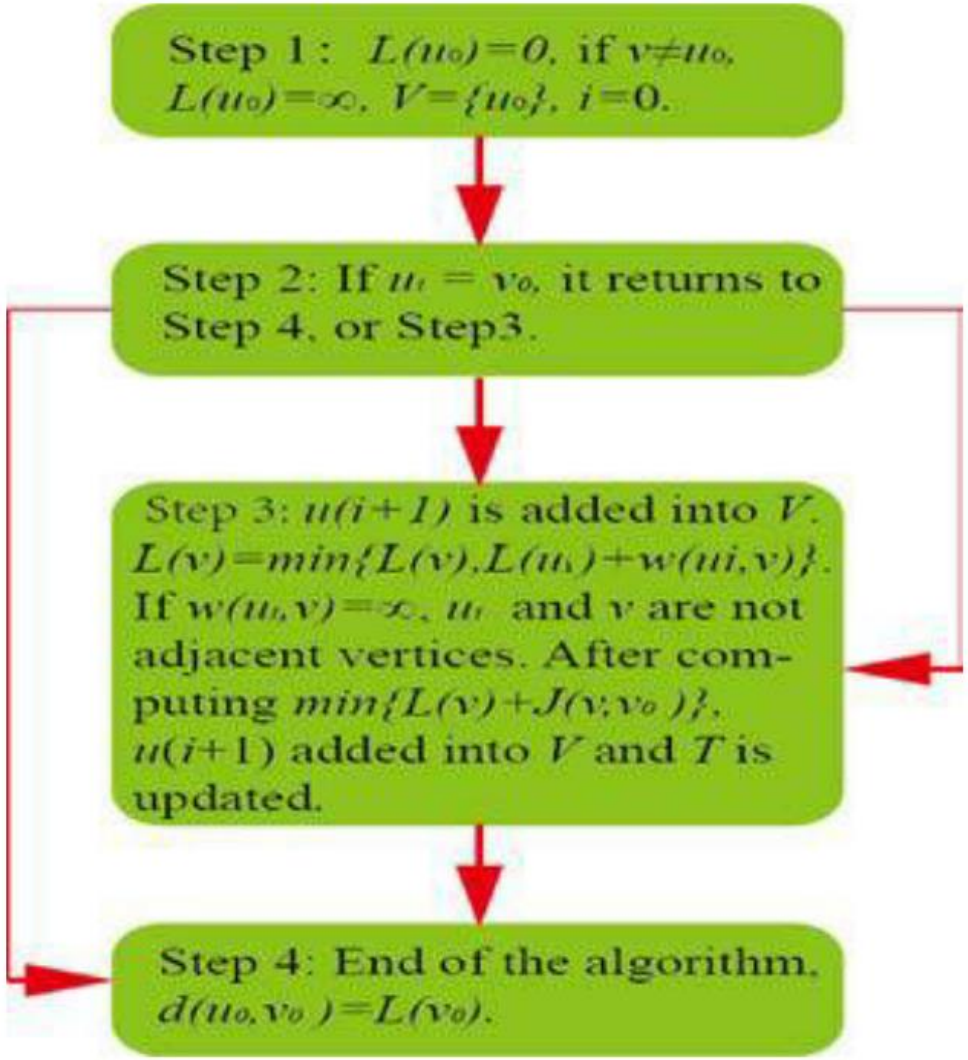
- **Dijkstra algorithm**

A type of search algorithm for identifying the shortest path between two points.

- Start point: u_0; Terminal point: v_0
- Collection: V(points of shortest path), T

- Dijkstra algorithm executes calculations of vertexes which have **nothing to do with the shortest path**. When there are much more points in T especially, the efficiency of the Dijkstra algorithm is **very low** due to overmuch computation.

2. Search of shortcut Dijkstra algorithm



$$L(u_{i+1}) + \mathcal{J}(u_{i+1}, v_0) = \min\{L(v) + \mathcal{J}(v, v_0)\}$$
$$(v \in T)$$

- Not select the vertex far from the shortest path
- Calculated amount is very small

More efficient than conventional Dijkstra algorithm

Fig.3-2 The procedure of the search of shortcut Dijkstra algorithm

Conventional APF algorithm

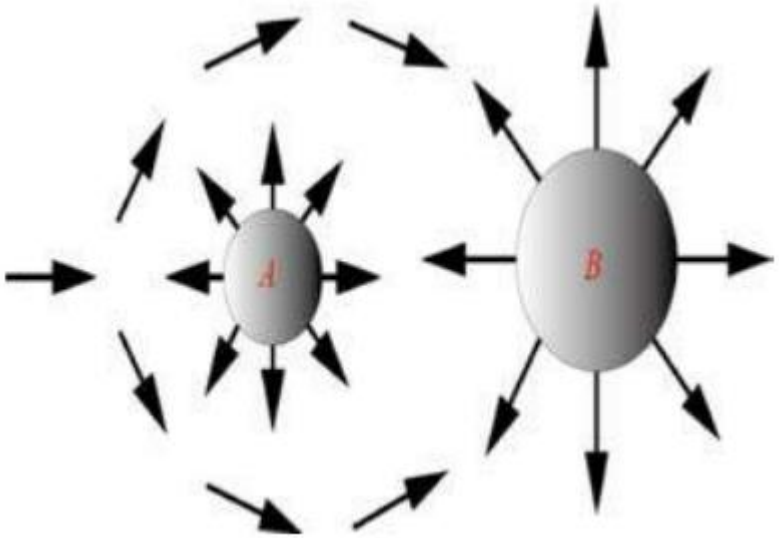


Fig.3-3 The model of APF:

A denotes obstacle, B is target

$$U_{art}(x) = U_{att}(x) + U_{rep}(x)$$

- Uart: APF

  Uatt: Attraction potential energy

  Urep: Repulsion potential energy

- **Resultant force**

$$F_{art} = F_{att} + F_{rep},$$
$$F_{att} = -grad[U_{att}(X)],$$
$$F_{rep} = -grad[U_{rep}(X)]$$

- GNRON problem

APF algorithm based on boundary detection of obstacles

- **Boundary detection**

Detects the next position with sensors when the robot runs into the region of obstacles to avoid **local minimum**, while a virtual force for escaping from this case is added.

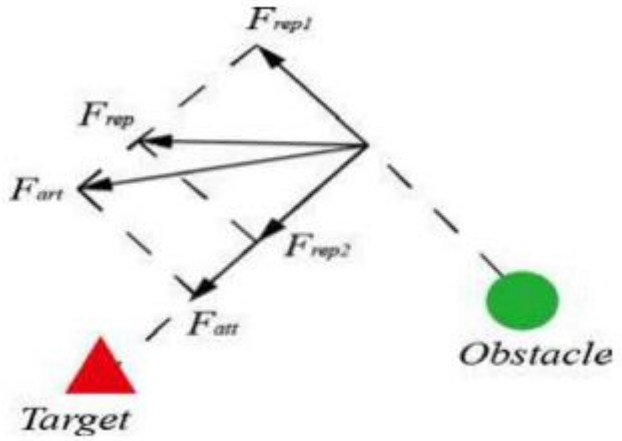


Fig.3-4 Artificial potential force diagram

- **3 conditions about local minimum**

$$A.\ |F_{att}(x) + F_{rep}(x)| \leq \varepsilon$$
$$B.\ |x(t+T) - x(t) \leq s_1|$$
$$C.\ |x_A - x_B| \leq \alpha \cdot s_{AB}$$

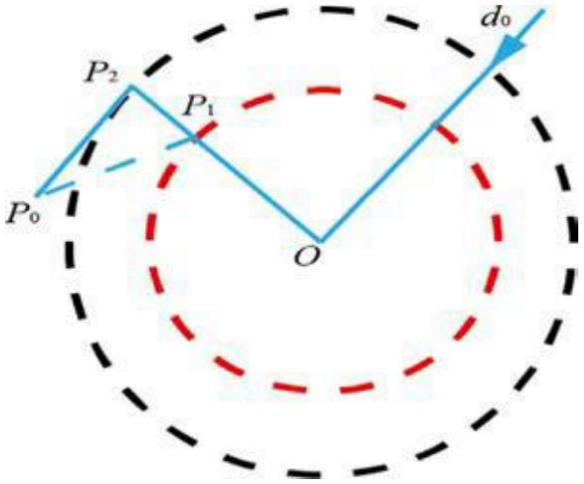


Fig.3-5 The diagram of boundary detection

O(x, y) : Center of an obstacle

d_0 : Its safety distance

P_0(x,y), P_1(x,y) : Original waypoint of the robot.

P_2(x,y) : Waypoint after revision of boundary detection.

$$|P_1(x,y) - O(x,y)| < d_0$$

θ: Direction angle between OP_1 and x-axis.

$$\theta = \arctan\left(\frac{O(x,y)_y - P_1(x,y)_y}{O(x,y)_x - P_1(x,y)_x}\right)$$

$$P_2(x,y)_x = O(x,y)_x - \cos\theta \cdot d_0$$
$$P_2(x,y)_y = O(x,y)_y + \sin\theta \cdot d_0$$

- **Secondary target**

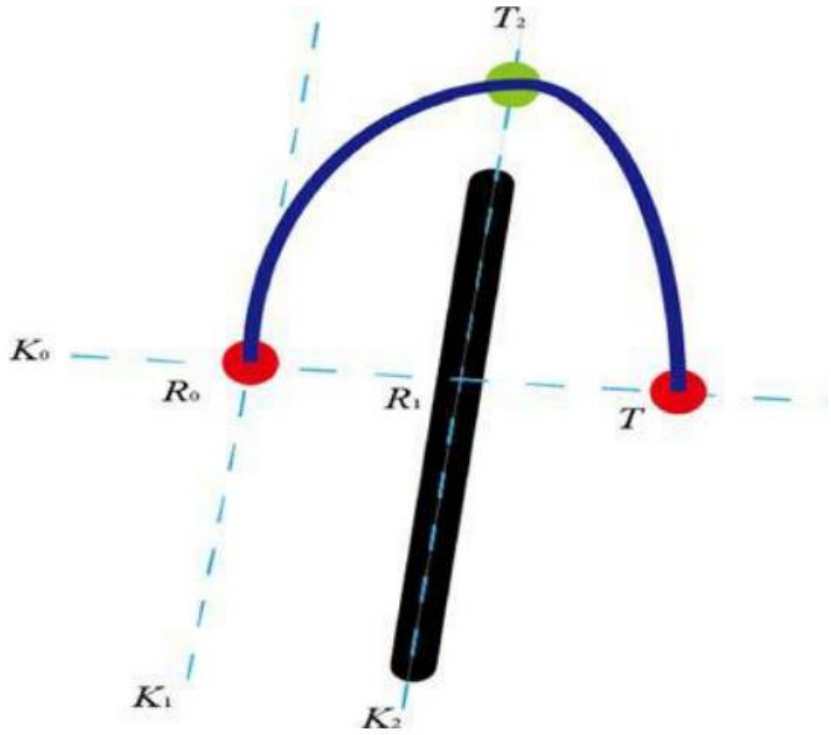


Fig.3-6 The diagram of setting secondary target

R_0: Current position of the robot

T : Target

T_2: Secondary target

- **How to set the secondary target?**

1. The straight-line equation K_0 of R_0 and T
2. The straight-line equation K_2
3. Straight-line equation K_1

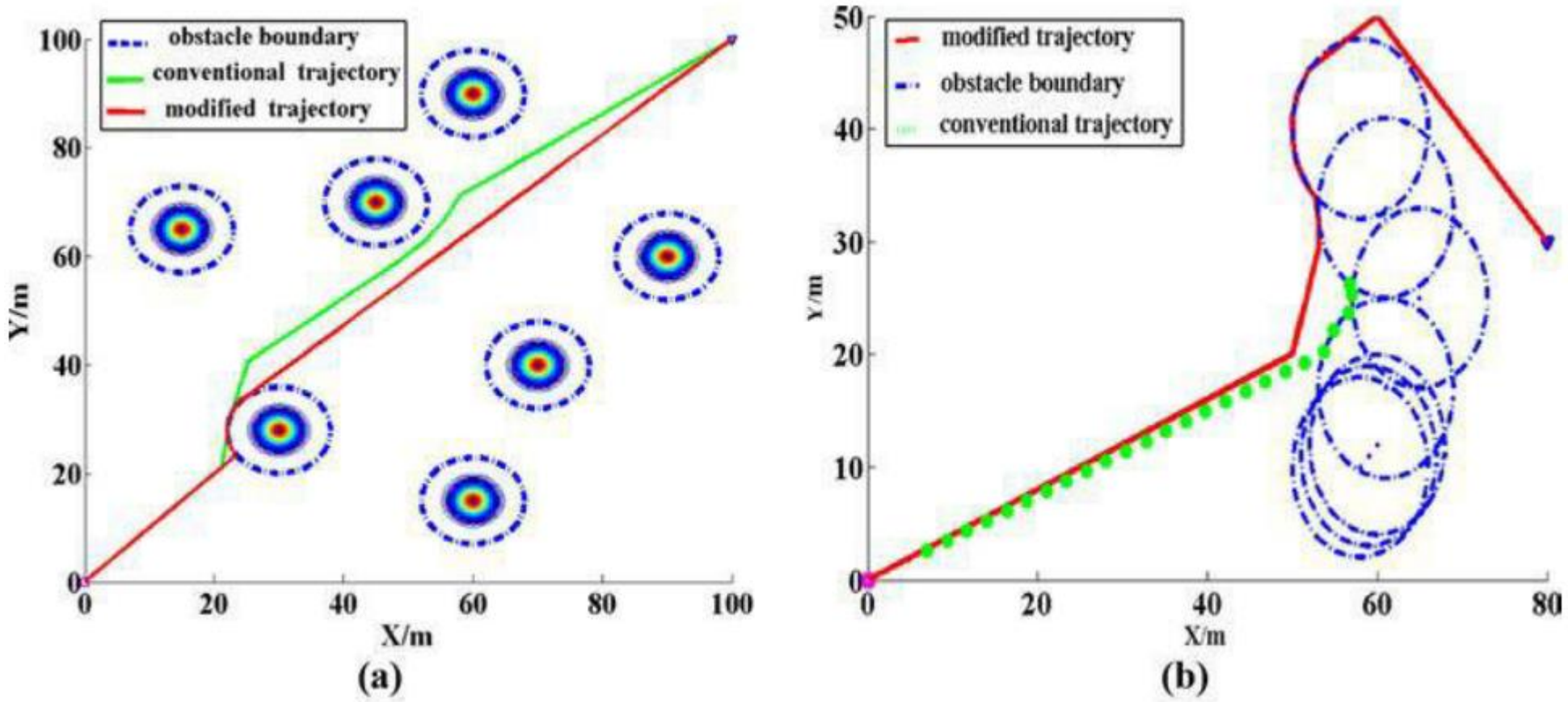


Fig.3-7 Simulation results in general case and local minimum

5. Experiments and results



Fig.3-8 Architecture of obstacle avoidance system



Fig.3-9 Obstacle-avoidance area

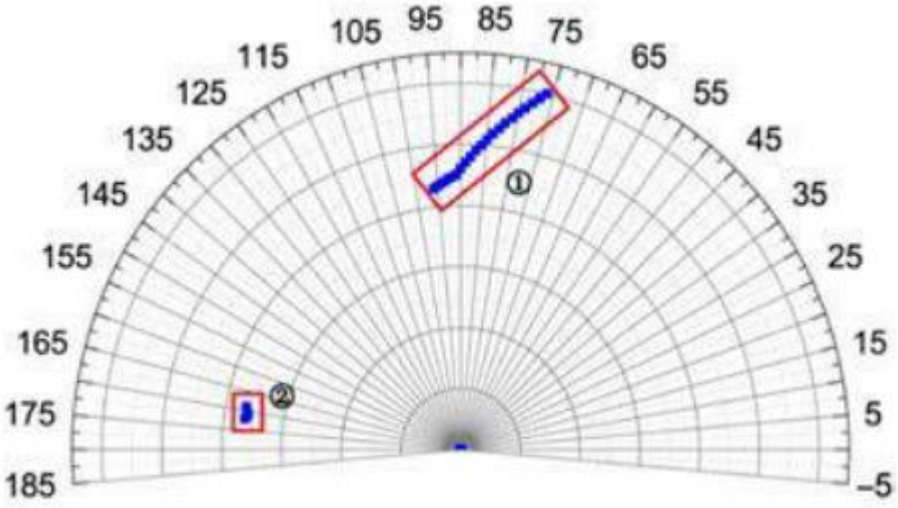


Fig.3-10 Obstacle detection results



Fig.3-11 The results of field test

6. Summary

- A search of **shortcut Dijkstra algorithm** is used in the global path planning.
- For USV navigation without prior knowledge, the **modified APF algorithm** is presented based on boundary detection in the local region of the specified map.
- The simulation and field tests show that the proposed algorithm can control the USV to avoid obstacles and return to the prior global path.
- **Future work:**
  - Obstacle avoidance algorithm
  - Obstacles detection
  - Trajectory tracking

THANKS

Tab.1-1 Properties of materials for an underwater robot

Specification	Density (kg/dm^3) (typical)	Tensile strength (MPa), Ultimate (typical)/Yield (typical)	Modulus of elasticity (GPa) (typical)
Steel (HY80)	7.87	482–689/276–517	190–210 (200)
Stainless steel (Type316)	7.75–8.1	620–560/415–290	193
Aluminum 6061-T6	2.7	310/276	68.9
Aluminum 7075-T6	2.81	572/503	71.7
Titanium Ti-6Al-4V	4.43	950/880	113.8
GFRP, E-glass	2.44–2.48	4800–4900/NA	85–90
GFRP, S-glass	2.54–2.60	3400–3500/NA	65–75
CFRP	1.75–1.95	2500–6000/NA	200–800
Acrylic	0.7–1.25 (1.16)	19.3–85.0 (64.3)/25.0–85.0 (60.2)	0.950–3.79 (2.69)
Polycarbonate	0.95–1.54 (1.2)	46.1–124 (64.6)/37.0–191 (64.3)	1.80–7.58 (2.37)
Unreinforced polyoxymethylene	1.14–2.23 (1.42)	5.00–115 (55.6)/22.0–120 (26.8)	0.586–12.1 (2.73)
Acrylonitrile–butadiene–styrene	0.35–3.5 (1.06)	24.1–73.1 (38)/20.0–73.1 (43.3)	0.778–6.10 (2.3)
Polyvinyl chloride	1.4	4.00–59.0/17.0–52.0	0.00159–3.24
Ceramic (96% alumina)	3.7–3.97	2070–2620/260–300	393
Metal matrix composite, Al/SiC	2.78–3.01	230–317	125–270